

Optimal Multidimensional Convolutional Codes

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Abstract

This paper analyzes m -dimensional (m D) convolutional codes with finite support in $\mathbb{N}^m$ over $\mathbb{F}_q[z_1, \dots, z_m]^n$, where $\mathbb{F}_q$ is a finite field with q elements. We revisit the m D generalized Singleton bound for codes with rate k/n and degree δ , focusing on convolutional codes that attain it, known as maximum distance separable (MDS) m D convolutional codes. As our main result, we develop new MDS m D convolutional codes constructions based on the superregularity of specific matrices. In particular, we provide new families of rate $1/n$ MDS m D codes by imposing row degree conditions on their generator matrices, significantly expanding the class of known MDS m D convolutional codes.

Key Words: MDS Codes, Multidimensional convolutional codes, Free distance.

1 Introduction

A one-dimensional (1D) convolutional code can be described as an $\mathbb{F}_q[z]$ -submodule of $\mathbb{F}_q[z]^n$, where $\mathbb{F}_q[z]$ denotes the polynomial ring in a single indeterminate over the field with q elements $\mathbb{F}_q$. Unlike block codes, their structure is ideal for continuous data streams, offering superior error correction by processing dependencies across multiple symbols. Fundamental theoretical foundations were established by Forney [For70, For73, For74], and the search for codes with optimal properties remains an active research field. For comprehensive introductions, see [JZ99, LPR21, LC83].

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Multidimensional (m D) convolutional codes extend the concept of convolutional codes to polynomial rings with multiple variables. An m D convolutional code of length n is a free R -submodule of R^n where $R = \mathbb{F}_q[z_1, \dots, z_m]$. Using m D codes for m -dimensional data instead of (several) codes of lower dimension, allows for low-delay encoding and decoding even if an entire $(m-1)$ -dimensional subset of the received word is erased. For instance, 2D convolutional codes are naturally suited for transmitting images and videos as 2D data. With the advancement of higher dimensional codes, we are confident that these applications will find more use in the future.

While 1D convolutional codes are well-studied, 2D codes are less explored. The concept of 2D convolutional codes was first introduced by Fornasini and Valcher in [VF94, FV94]. Subsequently, the authors in [CNPP16] established an upper bound for the free distance of 2D convolutional codes and proposed optimal code constructions. Further research on the construction and characterization of 2D convolutional codes can be found in [ANP18, ANP15, CNPP12]. Despite these advances, there are still relatively few explicit constructions of maximum distance separable (MDS) 2D convolutional codes. [CNPP12] proposed rate $1/n$ codes for $n \geq \frac{(\delta+1)(\delta+2)}{2}$, while [ANP15] achieved $n \geq \delta + 1$ for $\delta \leq 2$. For rate k/n , [CNPP16] proved the existence of MDS 2D convolutional codes if $n \geq \frac{k}{2}(\lfloor \frac{\delta}{k} \rfloor + 2)(\lfloor \frac{\delta}{k} \rfloor + 3)$ for $k \nmid \delta$, and $n \geq \frac{k}{2}(\lfloor \frac{\delta}{k} \rfloor + 1)(\lfloor \frac{\delta}{k} \rfloor + 2)$ for $k \mid \delta$. Recently, [ANP18] presented constructions satisfying $n \geq k(\lfloor \frac{\delta}{k} \rfloor + 1)$. Research on higher-dimensional (m D) convolutional codes is even more limited. First introduced in [Wei98, GLRW01] and further studied in [Cha09, CT04, Kit02, Zer10, Lom20], these codes exhibit fundamental differences from 1D and 2D cases [Wei98]. Very recently, [APS25] established the m D generalized Singleton bound and presented the first 3D MDS convolutional codes (rate $1/n$, $\delta \leq 2$) with finite support that achieve this bound. In addition to code constructions, decoding aspects have also been investigated, and in particular, a decoding algorithm for 2D convolutional codes over the erasure channel was proposed in [LP23].

In this paper, we develop new constructions of MDS m D convolutional codes of rate $1/n$. These constructions rely on the superregularity of certain matrices and extend known 1D results to the multidimensional setting.

2 Preliminaries

This section introduces the fundamental concepts of m D convolutional codes. Define the polynomial ring $R := \mathbb{F}_q[z_1, \dots, z_m]$.

Definition 1 ([Wei98]). *An m D (finite support) **convolutional code** $\mathcal{C}$ of rate k/n is a free R -submodule of R^n of rank k . A full row rank matrix $G(z_1, \dots, z_m) \in R^{k \times n}$, whose rows form a basis for $\mathcal{C}$, is called a **generator matrix** of $\mathcal{C}$ and therefore*

$$\mathcal{C} = \text{Im}_R G(z_1, \dots, z_m) = \{u(z_1, \dots, z_m)G(z_1, \dots, z_m) \text{ with } u(z_1, \dots, z_m) \in R^k\}.$$

Definition 2 ([Wei98]). A matrix $U(z_1, \dots, z_m) \in R^{k \times k}$ is **unimodular** if there is a matrix $V(z_1, \dots, z_m) \in R^{k \times k}$ such that $U(z_1, \dots, z_m) \cdot V(z_1, \dots, z_m) = V(z_1, \dots, z_m) \cdot U(z_1, \dots, z_m) = I_k$.

A matrix $U(z_1, \dots, z_m) \in R^{k \times k}$ is unimodular if and only if its determinant is in $\mathbb{F}_q \setminus \{0\}$ [Wei98]. Two full row rank matrices $G_1(z_1, \dots, z_m), G_2(z_1, \dots, z_m) \in R^{k \times n}$ are said to be **equivalent generator matrices** if they are generator matrices of the same m D convolutional code, i.e., if $\text{Im}_R G_1(z_1, \dots, z_m) = \text{Im}_R G_2(z_1, \dots, z_m)$, which happens if and only if $G_1(z_1, \dots, z_m) = U(z_1, \dots, z_m)G_2(z_1, \dots, z_m)$ for some unimodular matrix $U(z_1, \dots, z_m) \in R^{k \times k}$ [Wei98].

Let $\alpha = [\alpha_1 \dots \alpha_m] \in \mathbb{N}^m$. By z^α we mean $z_1^{\alpha_1} \dots z_m^{\alpha_m}$. A polynomial $f \in R$ can be written as $f(z_1, \dots, z_m) = \sum_{\alpha \in \mathbb{N}^m} f_\alpha z^\alpha$ where $f_\alpha \in \mathbb{F}_q$. In a similar way a vector $w = [w_1 \dots w_n] \in R^n$ can be written as $w = \sum_{\alpha \in \mathbb{N}^m} w_\alpha z^\alpha$ where $w_\alpha \in \mathbb{F}_q^n$.

Definition 3. The **weight** of $f \in R$ is denoted by $\text{wt}(f)$ and is the number of nonzero terms of f . If $w = [w_1 \dots w_n] \in R^n$, then the weight of w is given by $\text{wt}(w) = \sum_{j=1}^n \text{wt}(w_j)$. Equivalently, if $w = \sum_{\alpha \in \mathbb{N}^m} w_\alpha z^\alpha$ where $w_\alpha \in \mathbb{F}_q^n$, then $\text{wt}(w) = \sum_{\alpha \in \mathbb{N}^m} \text{wt}(w_\alpha)$.

Definition 4 ([Wei98]). Given two elements $w, \tilde{w} \in R^n$, the (Hamming) **distance** between w and $\tilde{w}$ is given by $d(w, \tilde{w}) = \text{wt}(w - \tilde{w})$. The **free distance** of an m D convolutional code $\mathcal{C} \subset R^n$ is given by $d_{\text{free}}(\mathcal{C}) = \min\{d(w, \tilde{w}) : w, \tilde{w} \in \mathcal{C}, w \neq \tilde{w}\}$.

The linearity of an m D convolutional code $\mathcal{C}$ implies that $d_{\text{free}}(\mathcal{C}) = \min\{\text{wt}(w) : w \in \mathcal{C}, w \neq 0\}$. The (total) degree of a polynomial $f = \sum_{\alpha \in \mathbb{N}^m} f_\alpha z^\alpha \in R$ is given by $\max\{\alpha_1 + \dots + \alpha_m : f_\alpha \neq 0\}$. The i -th row degree, ν_i , of a polynomial matrix $G(z_1, \dots, z_m) \in R^m$ is defined as the maximum degree of the entries in its i -th row. The **external degree** of $G(z_1, \dots, z_m)$ is the sum of its row degrees, i.e., $\sum_{i=1}^k \nu_i$.

Definition 5. Let $\mathcal{C}$ be an m D convolutional code. The **degree** of $\mathcal{C}$ is defined as the minimum of the external degrees among all the generator matrices of $\mathcal{C}$.

The following theorem, presented in [APS25], provides an upper bound on the distance of m D convolutional codes with rate k/n and degree δ , known as the **m D generalized Singleton bound**.

Theorem 1. Let $\mathcal{C}$ be an m D convolutional code of rate k/n and degree δ . Then $d_{\text{free}}(\mathcal{C}) \leq n \frac{(\lfloor \frac{\delta}{k} \rfloor + m)!}{\lfloor \frac{\delta}{k} \rfloor! m!} - k \left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1 \right) + \delta + 1$.

An mD convolutional code with rate k/n and degree δ whose free distance attains the corresponding mD generalized Singleton bound is called **Maximum Distance Separable (MDS) mD convolutional code**.

3 Construction of $1/n$ MDS multidimensional convolutional codes

In this section, we consider mD convolutional codes of rate $1/n$ and degree δ , and we present constructions of such codes that are MDS. The next definition defines superregular matrices which will be useful for the construction of these codes.

Definition 6. A matrix $A \in \mathbb{F}_q^{k \times n}$ is **superregular** if all of its minors (of all sizes) are nonzero.

Clearly any submatrix of a superregular matrix is also superregular and that a matrix obtained from a superregular matrix by permutation of rows is also superregular. The following lemma is well-known.

Lemma 2. Let $A \in \mathbb{F}_q^{r \times s}$ with $r \leq s$ be a superregular matrix. Then, for any $u \in \mathbb{F}_q^r \setminus \{0\}$, we have that $\text{wt}(uA) \geq s - \text{wt}(u) + 1$.

In [LP20] constructions of MDS 1D convolutional codes were presented using superregular matrices. Given a polynomial matrix $G(z) = \sum_{i=0}^{\delta} G_i z^i \in \mathbb{F}[z]^{1 \times n}$, with $G_{\delta} \neq 0$, let us consider the $(\delta + 1) \times n$ constant matrix

$$\Phi_1(G(z)) = \begin{bmatrix} G_0 \\ \vdots \\ G_{\delta} \end{bmatrix}. \quad (1)$$

Theorem 3. [LP20] Let $n, \delta \in \mathbb{N}$ with $n \geq \delta + 1$ and $G(z) = \sum_{i=0}^{\delta} G_i z^i \in \mathbb{F}[z]^{1 \times n}$, with $G_{\delta} \neq 0$. If $\Phi_1(G(z))$ is superregular, then $G(z)$ is a generator matrix of an MDS 1D convolutional code of rate $1/n$ and degree δ .

The following theorem gives an upper bound on the free distance of 1D convolutional codes whose generator matrices have specific row degrees, and provides a condition under which such codes achieve that upper bound.

Theorem 4. Let $\ell \in \mathbb{N}_0$ and $G_i(z) \in \mathbb{F}_q[z]^{k_i \times n}$ with $k_i \in \mathbb{N}$ for $i = 0, \dots, \ell - 1$ and $G_{\ell}(z) \in \mathbb{F}_q[z]^{1 \times n}$ such that $G(z) = \begin{bmatrix} G_0(z) \\ \vdots \\ G_{\ell}(z) \end{bmatrix} \in \mathbb{F}[z]^{(\sum_{s=0}^{\ell-1} k_s + 1) \times n}$ is the generator matrix of a convolutional code $\mathcal{C}$, where, for $i = 0, 1, \dots, \ell$,

all row degrees of $G_i(z)$ are equal to $\nu + \ell - i$ for some $\nu \in \mathbb{N}$. Then $d_{free}(\mathcal{C}) \leq n(\nu + 1)$.

Moreover, let $\mathcal{G} = \begin{bmatrix} \Phi_1(G_0(z)) \\ \Phi_1(G_1(z)) \\ \vdots \\ \Phi_1(G_\ell(z)) \end{bmatrix} \in \mathbb{F}^{L \times n}$ where $L = \sum_{j=0}^{\ell-1} k_j(\nu + \ell + 1 -$

$j) + \nu + 1$. If $n \geq L$ and $\mathcal{G}$ is superregular, then $d_{free}(\mathcal{C}) = n(\nu + 1)$.

Proof. First we observe that for $\ell = 0$ the statement of this theorem is exactly the statement of Theorem 3 and hence, for this proof we can assume $\ell \geq 1$.

Note that the codeword $G_\ell(z)$ has weight $n(\nu + 1)$ which proves that $d_{free}(\mathcal{C}) \leq n(\nu + 1)$. To prove that $d_{free}(\mathcal{C}) = n(\nu + 1)$ write $G(z) = G^{(0)} + G^{(1)}z + \dots + G^{(\nu+1)}z^{\nu+1} + \dots + G^{(\nu+\ell)}z^{\nu+\ell}$, where, for $j = 0, 1, \dots, \nu$, $G^{(j)}$ has $k_0 + k_1 + \dots + k_{\ell-1} + 1$ nonzero rows, and for $i = 1, 2, \dots, \ell$,

$$G^{(\nu+i)} = \begin{bmatrix} \tilde{G}^{(\nu+i)} \\ 0 \end{bmatrix} \quad \text{with} \quad \tilde{G}^{(\nu+i)} \in \mathbb{F}_q^{k_0+k_1+\dots+k_{\ell-i}}, \quad (2)$$

where all rows of $\tilde{G}^{(\nu+i)}$ are nonzero. Set $G^{(j)} = 0$, for $j > \nu + \ell$. Let us also write $u(z) = \sum_{i \in \mathbb{N}_0} u^{(i)}z^i$, with $u^{(i)} = \begin{bmatrix} u_0^{(i)} & u_1^{(i)} & \dots & u_{\ell-1}^{(i)} & u_\ell^{(i)} \end{bmatrix}$, where $u_j^{(i)} \in \mathbb{F}_q^{k_j}$, $j = 0, 1, \dots, \ell - 1$ and $u_\ell^{(i)} \in \mathbb{F}_q$. Set $u^{(j)} = 0$, for $j < 0$.

Let us prove that $w(z) = u(z)G(z)$ has weight greater or equal than $n(\nu + 1)$. We can consider without loss of generality that $u^{(0)} \neq 0$. For $w(z) = \sum_{i \in \mathbb{N}_0} w_i z^i$, let us denote $w|_{[r,s]} = \sum_{i=r}^s w_i z^i$, for $(r < s)$. Since $\mathcal{G}$ is a

superregular matrix and $u^{(0)} \neq 0$, we have that $w(z)$ has degree greater or equal than ν .

Case 1: $\deg w(z) = \nu$.

Then,

$$w^{(\nu+1)} = \begin{bmatrix} u^{(\nu+1)} & \dots & u^{(0)} \end{bmatrix} \begin{bmatrix} G^{(0)} \\ \vdots \\ G^{(\nu+1)} \end{bmatrix} = 0. \quad (3)$$

Recall that for $j = 0, \dots, \nu$, all rows of $G^{(j)}$ are nonzero and $G^{(\nu+1)}$ has exactly the last row equal to zero since we can assume $\ell \geq 1$. After deleting this zero row, the matrix in (3) is a full row rank submatrix of $\mathcal{G}$ since $\mathcal{G}$ is superregular with at least as many columns as rows. Hence, (3) implies that $u^{(\nu+1)} = u^{(\nu)} = \dots = u^{(1)} = 0$ and $u_j^{(0)} = 0$ for $j = 0, 1, \dots, \ell - 1$. Moreover, it can be easily seen that $u^{(\nu+i)} = 0$ for $i \geq 2$, i.e., $u(z) = \begin{bmatrix} 0 & \dots & 0 & \bar{u} \end{bmatrix}$ with $\bar{u} \in \mathbb{F}_q \setminus \{0\}$. Thus, $w(z) = \bar{u} G_\ell(z)$, which has weight $n(\nu + 1)$.

Case 2: $\deg w(z) = \nu + i$ for some $i \geq 1$.

Then, $w^{(\nu+i+1)} = \begin{bmatrix} u^{(\nu+i+1)} & u^{(\nu+i)} & \dots & u^{(i+1-\ell)} \end{bmatrix} \begin{bmatrix} G^{(0)} \\ \vdots \\ G^{(\nu+\ell)} \end{bmatrix} = 0$, which

implies that

$$u^{(\nu+i+1)} = u^{(\nu+i)} = \dots = u^{(i+1)} = 0. \quad (4)$$

Moreover, since $w^{(\nu+i+j)} = 0$, for $j > 2$, it follows that $u^{(\nu+i+j)} = 0$, for $j > 2$, which means that $\deg u(z) = i$. Furthermore, for $s = 0, \dots, \ell - 1$,

$$u^{(i-s)} = \begin{bmatrix} 0 & \bar{u}^{(i-s)} \end{bmatrix} \quad \text{where} \quad \bar{u}^{(i-s)} \in \mathbb{F}_q^{1+k_{\ell-1}+\dots+k_{\ell-s}}, \quad (5)$$

since $G^{(\nu+1+s)}$ has $k_0 + k_1 + \dots + k_{\ell-1-s}$ nonzero rows; see (2).

Then, for $j = 0, 1, \dots, \min\{i-1, \nu\}$,

$$\begin{aligned} w^{(\nu+i-j)} &= \begin{bmatrix} u^{(i)} & \dots & u^{(i+1-\ell)} & u^{(i-\ell)} & \dots & u^{(i-\ell-j)} \end{bmatrix} \begin{bmatrix} G^{(\nu-j)} \\ \vdots \\ G^{(\nu+\ell)} \end{bmatrix} \\ &= \begin{bmatrix} \bar{u}^{(i)} & \dots & \bar{u}^{(i+1-\ell)} & u^{(i-\ell)} & \dots & u^{(i-\ell-j)} \end{bmatrix} \begin{bmatrix} \bar{G}^{(\nu-j)} \\ \vdots \\ \bar{G}^{(\nu-j-1+\ell)} \\ G^{(\nu-j+\ell)} \\ \vdots \\ G^{(\nu+\ell)} \end{bmatrix} \end{aligned}$$

where $G^{(\nu-j+s)} = \begin{bmatrix} \hat{G}^{(\nu-j+s)} \\ \bar{G}^{(\nu-j+s)} \end{bmatrix}$ for $s = 0, 1, \dots, \ell - 1$, with $\bar{G}^{(\nu-j)} \in \mathbb{F}_q^{1 \times n}$ and $\bar{G}^{(\nu-j+s)} \in \mathbb{F}_q^{(1+k_{\ell-1}+\dots+k_{\ell-s}) \times n}$.

Since the number of rows of the matrix $\begin{bmatrix} \bar{G}^{(\nu-j)} \\ \vdots \\ \bar{G}^{(\nu-j+\ell-1)} \\ G^{(\nu-j+\ell)} \\ \vdots \\ G^{(\nu+\ell)} \end{bmatrix}$ is

$$\begin{aligned} &\sum_{s=0}^{\ell-1} (1 + k_{\ell-1} + \dots + k_{\ell-s}) + \sum_{i=0}^j (k_0 + \dots + k_i) \\ &= \sum_{i=0}^{\ell-1} (1 + k_{\ell-1} + \dots + k_{i+1}) + \sum_{i=0}^j (k_i + \dots + k_0) \\ &= \begin{cases} (j+1)(k_0 + \dots + k_{\ell-1} + 1) + \sum_{i=j+1}^{\ell-1} (1 + k_{\ell-1} + \dots + k_{i+1}), & j \leq \ell - 1 \\ \ell(k_0 + \dots + k_{\ell-1} + 1) + \sum_{i=\ell}^j (k_i + \dots + k_0), & j > \ell - 1 \end{cases} \end{aligned}$$

where $k_\ell = 1$ and $k_t = 0$ for $t > \ell$, there are at least $(j+1)(k_0 + k_1 + \dots + k_{\ell-1} + 1)$ nonzero rows which are a submatrix of $\mathcal{G}$. Moreover, $\deg(u(z)) = i$

implies $\bar{u}^{(i)} \neq 0$ and thus, by Lemma 2, we have that

$$\text{wt}(w^{(\nu+i-j)}) \geq n - (j+1)(k_0 + k_1 + \cdots + k_{\ell-1} + 1) + 1. \quad (6)$$

Case 2.1: $i \geq \nu + 1$

In this case, (6) is true for $j = 0, \dots, \nu$ and we obtain $\text{wt}(w_{[i, \nu+i]}) = \sum_{j=0}^{\nu} \text{wt}(w^{(\nu+i-j)}) \geq n(\nu+1) - \frac{(\nu+1)(\nu+2)}{2} (k_0 + \cdots + k_{\ell-1} + 1) + \nu + 1$.

Moreover, we have that for $t = 0, \dots, \nu$,

$$w^{(t)} = \begin{bmatrix} u^{(t)} & \cdots & u^{(0)} \end{bmatrix} \begin{bmatrix} G^{(0)} \\ \vdots \\ G^{(t)} \end{bmatrix} \quad (7)$$

has weight greater than or equal to $n - (t+1)(k_0 + k_1 + \cdots + k_{\ell-1} + 1) + 1$, due to Lemma 2 and therefore,

$$\text{wt}(w_{[0, \nu]}) \geq n(\nu+1) - \frac{(\nu+1)(\nu+2)}{2} (k_0 + k_1 + \cdots + k_{\ell-1} + 1) + \nu + 1.$$

In sum, one obtains

$$\begin{aligned} \text{wt}(w(z)) &\geq \text{wt}(w_{[0, \nu]}) + \text{wt}(w_{[i, \nu+i]}) \\ &\geq n(\nu+1) + n(\nu+1) - (\nu+1)(\nu+2)(k_0 + \cdots + k_{\ell-1} + 1) + 2(\nu+1) \\ &\geq n(\nu+1) + (\nu+1)(\nu+1 - (\nu+2) + 2) \geq n(\nu+1), \end{aligned}$$

where the third inequality follows from $n \geq \sum_{j=0}^{\ell-1} k_j(\nu + \ell + 1 - j) + \nu + 1 \geq (\nu+2) \sum_{j=0}^{\ell-1} k_j + \nu + 1$.

Case 2.2: $i < \nu + 1$

In this case, we can use (6) for $j = 0, \dots, i-1$ to obtain

$$\text{wt}(w_{[\nu+1, \nu+i]}) \geq ni - \frac{i(i+1)}{2} (k_0 + k_1 + \cdots + k_{\ell-1} + 1) + i.$$

Case 2.2.1: $i < \nu + 1$ and $i \geq \ell - 1$

In this case, using again Lemma 2, we obtain

$$\text{wt}(w^{(t)}) \geq n - (t+1)(k_0 + k_1 + \cdots + k_{\ell-1} + 1) + 1, \text{ for } t = 0, 1, \dots, i - \ell,$$

For $t \geq i - \ell + 1$, we can exploit that according to (5), for $s = 0, \dots, \ell - 1$, the vector $u^{(i-s)}$ has (at least) $k_0 + \cdots + k_{\ell-s-1}$ zero components. Hence, considering again (7) and observing that for $t \geq i - \ell + 1$, the vectors $u^{(t)}, \dots, u^{(i-\ell+1)}$ are involved in the calculation of $w^{(t)}$, we obtain

$$\text{wt}(w^{(t)}) \geq n - (t+1) \left(\sum_{s=0}^{\ell-1} k_s + 1 \right) + 1 + \sum_{j=0}^{\ell-i+t-1} (\ell - i + t - j) k_j$$

for $t = i - \ell + 1, \dots, i$ and

$$\text{wt}(w^{(t)}) \geq n - (i+1)(k_0 + k_1 + \cdots + k_{\ell-1} + 1) + 1 + \ell k_0 + (\ell-1)k_1 + \cdots + k_{\ell-1},$$

for $t = i, i + 1, \dots, \nu$. Therefore,

$$\begin{aligned} \text{wt}(w|_{[0, i-1]}) &\geq ni - \frac{i(i+1)}{2} \left(\sum_{s=0}^{\ell-1} k_s + 1 \right) + i + \sum_{j=0}^{\ell-1} \frac{(j+1)(j+2)}{2} k_{\ell-1-j}, \\ \text{wt}(w|_{[i, \nu]}) &\geq n(\nu - i + 1) - (\nu - i + 1)(i + 1)(k_0 + k_1 + \dots + k_{\ell-1} + 1) + \\ &\quad + \nu - i + 1 + (\nu - i + 1)(\ell k_0 + (\ell - 1)k_1 + \dots + k_{\ell-1}). \end{aligned}$$

Thus, since $n \geq L = \sum_{j=0}^{\ell-1} k_j(\nu + \ell + 1 - j) + \nu + 1$, we have that

$$\begin{aligned} \text{wt}(w(z)) &= \text{wt}(w|_{[0, \nu]}) + \text{wt}(w|_{[\nu+1, \nu+i]}) \\ &\geq n(\nu + 1) + ni - (i + 1)(\nu + 1) \left(\sum_{j=0}^{\ell-1} k_0 + 1 \right) \\ &\quad + \nu + 1 + i + \sum_{j=0}^{\ell-1} \left(\frac{(j+1)(j+2)}{2} + (j+1)(\nu - i + 1) \right) k_{\ell-1-j} \\ &\geq n(\nu + 1) + \sum_{j=0}^{\ell-1} \left(i(\ell - j - 1) + \frac{(j+1)(j+2)}{2} + j(\nu - i + 1) \right) k_{\ell-1-j} \\ &\geq n(\nu + 1). \end{aligned}$$

Case 2.2.2: $i < \nu + 1$ and $i < \ell - 1$

In this case, we have that $w^{(\nu+i+1)} = \begin{bmatrix} u^{(i)} & \dots & u^{(0)} \end{bmatrix} \begin{bmatrix} G^{(\nu+1)} \\ \vdots \\ G^{(\nu+i+1)} \end{bmatrix} = 0$ implies that $u(z) = \begin{bmatrix} 0 & \dots & 0 & \hat{u}(z) \end{bmatrix}$ with $\hat{u}(z) \in \mathbb{F}_q[z]^{k_{\ell-i} + \dots + k_{\ell-1} + 1}$, and therefore $w(z) = \hat{u}(z) \begin{bmatrix} G_{\ell-i}(z) \\ \vdots \\ G_{\ell}(z) \end{bmatrix}$. Setting $\hat{\ell} = i$ and $\hat{k}_j = k_{j+\ell-i}$ for $j = 0, \dots, i$, i.e. in particular $\hat{\ell} \geq i$, brings us into the situation of Case 2.2.1. Hence, we obtain $\text{wt}(w(z)) \geq n(\nu + 1)$. $\square$

Let us now consider m -dimensional convolutional codes. Next, we introduce the following technical lemmas, which will be used in the proof of the subsequent theorem. The second of these lemmas is also known as hockey stick identity.

Lemma 5 ([Ros02]). *Let $m, \nu \in \mathbb{N}$ and $S = \{(i_1, i_2, \dots, i_m) : 0 \leq i_1 + i_2 + \dots + i_m \leq \nu\}$. Then $\#S = \frac{(\nu+m)!}{\nu!m!}$.*

Lemma 6 ([ALPS26]). *Let $m, \nu \in \mathbb{N}$. Then $\frac{(\nu+m)!}{\nu!m!} = \sum_{i=0}^{\nu} \frac{(\nu+m-i-1)!}{(\nu-i)!(m-1)!}$*

Given a polynomial matrix $G(z_1, z_2, \dots, z_m) = \sum_{i=0}^{\mu} G_i(z_1, \dots, z_{m-1})z_m^i \in R^{1 \times n}$, with $G_{\mu}(z_1, \dots, z_{m-1}) \neq 0$, let us define the constant matrix

$$\Phi_m(G(z_1, z_2, \dots, z_m)) = \begin{bmatrix} \Phi_{m-1}(G_0(z_1, \dots, z_{m-1})) \\ \Phi_{m-1}(G_1(z_1, \dots, z_{m-1})) \\ \vdots \\ \Phi_{m-1}(G_{\mu}(z_1, \dots, z_{m-1})) \end{bmatrix}, \text{ for } m \geq 2,$$

with $\Phi_1(G_i(z_1))$ as defined in (1).

Theorem 7. Let $G(z_1, \dots, z_m) = \begin{bmatrix} G_0(z_1, \dots, z_m) \\ \vdots \\ G_{\ell}(z_1, \dots, z_m) \end{bmatrix} \in R^{(k_0 + \dots + k_{\ell-1} + 1) \times n}$

be a generator matrix of an mD convolutional code $\mathcal{C}$, with $G_0(z_1, \dots, z_m) \in R^{k_0 \times n}$, $G_1(z_1, \dots, z_m) \in R^{k_1 \times n}$, $\dots$, $G_{\ell-1}(z_1, \dots, z_m) \in R^{k_{\ell-1} \times n}$, $G_{\ell}(z_1, z_2, \dots, z_m) \in R^{1 \times n}$ of degrees $\nu_0 \geq \nu_1 \geq \dots \geq \nu_{\ell-1} > \nu_{\ell}$, respectively. Then, $d_{\text{free}}(\mathcal{C}) \leq n \frac{(\nu_{\ell} + m)!}{\nu_{\ell}! m!}$. Moreover, if $n \geq k_0(\nu_0 + 1) + k_1(\nu_1 + 1) + \dots + k_{\ell-1}(\nu_{\ell-1} + 1) + \nu_{\ell} + 1$ and $\Phi_m(G(z_1, z_2, \dots, z_m))$ is superregular, then $d_{\text{free}}(\mathcal{C}) = n \frac{(\nu_{\ell} + m)!}{\nu_{\ell}! m!}$.

Proof. Let $w(z_1, \dots, z_m) = [0 \dots 0 \ 1] \begin{bmatrix} G_0(z_1, \dots, z_m) \\ \vdots \\ G_{\ell}(z_1, \dots, z_m) \end{bmatrix} = G_{\ell}(z_1, \dots, z_m)$. By

Lemma 5, $d_{\text{free}}(\mathcal{C}) \leq \text{wt}(G_{\ell}(z_1, \dots, z_m)) \leq n \cdot \# \text{Supp}(G_{\ell}(z_1, \dots, z_m)) = n \frac{(\nu_{\ell} + m)!}{\nu_{\ell}! m!}$, where the support of $G(z_1, \dots, z_m) = \sum_{i_1, \dots, i_m \in \mathbb{N}} G_{i_1, \dots, i_m} z_1^{i_1} \dots z_m^{i_m} \in R^m$ is $\text{Supp}(G(z_1, \dots, z_m)) = \{(i_1, i_2, \dots, i_m) \in \mathbb{N}^m : G_{i_1, i_2, \dots, i_m} \neq 0\}$. Let

us assume that $\Phi_m(G(z_1, \dots, z_m))$ is superregular and let us prove that

$$d_{\text{free}}(\mathcal{C}) = n \frac{(\nu_{\ell} + m)!}{\nu_{\ell}! m!} \quad (8)$$

by induction. By Theorem 4, (8) is valid for $m = 1$. Let us now consider that (8) is valid for $m - 1$, i.e., let us consider as induction hypothesis that

if $\tilde{G}(z_1, \dots, z_{m-1}) = \begin{bmatrix} \tilde{G}_0(z_1, \dots, z_{m-1}) \\ \vdots \\ \tilde{G}_{\tilde{\ell}}(z_1, \dots, z_{m-1}) \end{bmatrix} \in \mathbb{F}_q[z_1, \dots, z_{m-1}]^{(\tilde{\ell}+1) \times n}$ is a gen-

erator matrix of an $(m-1)D$ convolutional code $\tilde{\mathcal{C}}$, with $\tilde{G}_0(z_1, \dots, z_{m-1})$, $\tilde{G}_1(z_1, \dots, z_{m-1})$, $\dots$, $\tilde{G}_{\tilde{\ell}}(z_1, \dots, z_{m-1})$ of degrees $\tilde{\nu}_0 \geq \tilde{\nu}_1 \geq \dots \geq \tilde{\nu}_{\tilde{\ell}-1} > \tilde{\nu}_{\tilde{\ell}}$, respectively, and $\Phi_{m-1}(\tilde{G}(z_1, \dots, z_m))$ is superregular, then $d_{\text{free}}(\tilde{\mathcal{C}}) = n \frac{(\tilde{\nu}_{\tilde{\ell}} + m - 1)!}{\tilde{\nu}_{\tilde{\ell}}! (m-1)!}$. Write

$$\begin{aligned} G(z_1, \dots, z_m) &= G^{(0)}(z_1, \dots, z_{m-1}) + G^{(1)}(z_1, \dots, z_{m-1})z_m + \dots \\ &\quad + G^{(\nu_{\ell})}(z_1, \dots, z_{m-1})z_m^{\nu_{\ell}} + \dots + G^{(\nu_0)}(z_1, \dots, z_{m-1})z_m^{\nu_0}. \end{aligned}$$

Note that $G^{(i)}(z_1, \dots, z_{m-1})$ has row degrees $\nu_0 - i, \nu_1 - i, \dots, \nu_\ell - i$, for $i = 0, \dots, \nu_\ell$. Let us consider $u(z_1, \dots, z_m) \in R^{\ell+1} \setminus \{0\}$ and write

$$\begin{aligned} u(z_1, \dots, z_m) \\ = u^{(0)}(z_1, \dots, z_{m-1}) + u^{(1)}(z_1, \dots, z_{m-1})z_m + \dots + u^{(t)}(z_1, \dots, z_{m-1})z_m^t, \end{aligned}$$

for some $t \in \mathbb{N}$. Let $w(z_1, \dots, z_m) = u(z_1, \dots, z_{m-1})G(z_1, \dots, z_m)$, and let us prove that $\text{wt}(w(z_1, z_2, \dots, z_m)) \geq n \frac{(\nu_\ell + m)!}{\nu_\ell! m!}$.

Without loss of generality, we assume $u^{(0)}(z_1, \dots, z_{m-1}) \neq 0$. Write $w(z_1, z_2, \dots, z_m) = \sum_{i \in \mathbb{N}} w^{(i)}(z_1, \dots, z_{m-1}) z_m^i$, where, for $i = 0, 1, \dots, \nu_\ell$

$$\begin{aligned} w^{(i)}(z_1, \dots, z_{m-1}) \\ = [u^{(i)}(z_1, \dots, z_{m-1}) \quad \dots \quad u^{(0)}(z_1, \dots, z_{m-1})] \times \begin{bmatrix} G^{(0)}(z_1, \dots, z_{m-1}) \\ \vdots \\ G^{(i)}(z_1, \dots, z_{m-1}) \end{bmatrix} z_m^i, \end{aligned}$$

As $\Phi_m(G(z_1, \dots, z_m))$ is superregular, all matrices $\Phi_{m-1}([G^{(0)}(z_1, \dots, z_{m-1})])$,

$$\Phi_{m-1}\left(\begin{bmatrix} G^{(0)}(z_1, \dots, z_{m-1}) \\ G^{(1)}(z_1, \dots, z_{m-1}) \end{bmatrix}\right), \dots, \Phi_{m-1}\left(\begin{bmatrix} G^{(0)}(z_1, \dots, z_{m-1}) \\ \vdots \\ G^{(\nu_\ell)}(z_1, \dots, z_{m-1}) \end{bmatrix}\right)$$

are also superregular. Moreover, the matrices $G^{(0)}(z_1, \dots, z_{m-1})$, $\begin{bmatrix} G^{(0)}(z_1, \dots, z_{m-1}) \\ G^{(1)}(z_1, \dots, z_{m-1}) \end{bmatrix}$, $\dots$,

$\begin{bmatrix} G^{(0)}(z_1, \dots, z_{m-1}) \\ \vdots \\ G^{(\nu_\ell)}(z_1, \dots, z_{m-1}) \end{bmatrix}$ have the last row of degrees $\nu_\ell, \nu_\ell - 1, \dots, 0$, respectively,

and the remaining rows of each matrix have higher degree than their last row. Thus, by induction hypothesis, $\text{wt}(w^{(i)}(z_1, \dots, z_{m-1})) \geq n \frac{(\nu_\ell + m - i - 1)!}{(\nu_\ell - i)!(m-1)!}$, $i = 0, 1, \dots, \nu_\ell$. Therefore, by Lemma 6, $\text{wt}(w(z_1, \dots, z_m)) \geq n \frac{(\nu_\ell + m)!}{\nu_\ell! m!}$. $\square$

The next theorem follows immediately from Theorem 7 and it gives a condition for the existence (and construction) of MDS mD convolutional codes of rate $\frac{1}{n}$.

Theorem 8. *Let $G(z_1, \dots, z_m) \in R^{1 \times n}$ with row degree equal to δ and $n \geq \delta + 1$, such that $\Phi_m(G(z_1, \dots, z_m))$ is superregular. Then $G(z_1, \dots, z_m)$ is a generator matrix of an MDS mD convolutional code of rate $1/n$ and degree δ .*

Finally, note that the existence of such codes is closely related to the existence of the superregular matrices introduced above.

Moreover, note that with the help of Theorem 7 one can also construct MDS mD convolutional codes of general rate k/n and degree $\delta = k\nu + k - 1$ with specific requirements on the row degrees, see Theorem 3.8 of [ALPS26].

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